Algorithm:

1. Create a MusicLoop class using a Circular Linked List.
2. Initialize the last pointer to nullptr.
3. Add songs to the playlist by creating new nodes.
4. Connect the last node to the first node to form a Circular Linked List.
5. Display the title “Music Loop System”.
6. Check whether the playlist is empty.
7. If the playlist is empty or the number of loops is invalid, display an appropriate message.
8. Copy the songs from the Circular Linked List into a vector.
9. Shuffle the songs randomly for each loop.
10. Display the songs in their shuffled order.
11. Repeat the process until the specified number of loops is completed.
12. Delete all dynamically allocated nodes.
13. Stop the program.